

FreeMoCap X-Lab

Lead Institution Name

Written Proposal to the NSF X-Labs Initiative

Topic: Scientific Instrumentation for Sensing and Imaging

1. Mission

Proposing teams must describe one or more novel platform technologies that the proposed NSF X-Lab is uniquely suited to advance. Successful teams will have a clear vision statement of how their work will result in an end-state that reshapes an entire field of scientific inquiry or a sector of technology, and how their Mission is currently unmet by existing organizational structures and funding mechanisms.

2. Technology Landscape

Provide an analysis of the current technology landscape that justifies how the proposed platform technology would accelerate the emergence of new technologies and lines of research. The Technology Landscape Analysis should compare current state-of-the-art technology and should provide a clear description of the team's goal to progress the technology and unlock fundamentally new capabilities. A plot or other figure to quantitatively depict the current technology landscape and desired goal(s) of the NSF X-Lab may be included here.

3. Outcomes

An NSF X-Lab is expected to deliver high-impact, measurable outcomes that advance a nascent platform technology, overcome critical bottlenecks, and realize the team's vision to reshape an entire area of technology or field of scientific research (see Section 7.0 for example outcomes). Phase 0 proposals should describe the team's ambitious target outcomes over a 5- to 7-year timescale to accomplish the stated Mission and the scientific and technical performance benchmarks needed to get there.

4. Senior/Key Personnel Qualifications

Provide a brief description of the qualifications of Senior/Key Personnel on the team. Include technical expertise, experience commercializing technology, past efforts, and other qualifications that will contribute to achieving the team's Mission. Include information for all Senior/Key Personnel, including team members not currently affiliated with the lead organization.

5. Team Capabilities Statement

Provide a Team Capability Statement describing how the Senior/Key Personnel on the proposed leadership team bring together complementary expertise in strategic leadership, technical expertise, and/or entrepreneurship, as applicable. The statement should describe how, collectively, the team demonstrates the experience, adaptability, and collaborative approach necessary to execute the Mission. Briefly describe the team's intended governance

structure and autonomy during Phase 0 and Phase 1. If applicable, include an overview of networks, partnerships, and capital resources your team currently has and how they will be deployed in support of the proposal.